

My research interests are in designing scalable algorithms for large and high-dimensional datasets. Such datasets routinely arise in big data analysis and machine learning, where large collections of real-world objects (like images, videos or text) are mapped to vectors of many numerical features, viewed as points in a high-dimensional space. This endows the data with a *metric structure*, that forms the basis of subsequent processing (such as searching, clustering or classification). Unfortunately, the sheer size of modern datasets – billions of points with hundreds of features are common – renders many traditional algorithms infeasible in practice. This requires new algorithmic techniques that can work at a much larger scale.

My work chiefly relies on *compact representations of metric data*. In this approach, the data is summarized into a short synopsis that captures its key properties, and on which computation can be performed much more efficiently. This powerful paradigm is prevalent across all areas of computing, under various nomenclatures – *compression*, *sketching*, *hashing*, *quantization*, and *sparsification*, to name a few – with far-reaching applications from fundamental theory to cutting-edge systems. I am especially interested in those techniques that are implementable and practical, and can lead to real-world impact.

Overview of Main Contributions

Optimal metric compression. Perhaps the most fundamental question on metric data compression is: What is the minimal number of bits required to store all distances between data points, up to a prescribed precision? In my core thesis research, I obtained tight upper and lower bounds on the bit representation size of Euclidean metric spaces. I extended the result to support Nearest Neighbor Search queries, achieving the smallest known data structure for this problem with worst-case guarantees. These results mark the first improvement to the storage size of Euclidean metrics over classic dimensionality reduction.

The underlying compression technique is general and robust, and has yielded several additional results, including tight bounds for general metric spaces. Furthermore, the method is practical: I have implemented it and shown it empirically matches or improves state-of-the-art performance.

Fast algorithms via metric compression. Compact metric representations ostensibly improve the storage size of the data – but are also useful for directly improving running times. In this vein, my work described below has contributed algorithms for semi-supervised learning on data streams, scalable fair clustering, and nearest neighbor search assisted by neural networks. My other works include fast approximation algorithms for optimal transport, kernel density estimation in high dimensions, and low-rank approximation of distance matrices. All of these algorithms have been implemented and shown to outperform previous state-of-the-art. I have also introduced the by now well-studied notion of *resistance sparsifiers*, analyzed widely used heuristics for massive eccentricity computation, and studied combinatorial notions of multitasking that arise in cognitive science.

Optimal Metric Compression

Contemporary datasets are almost invariably represented as high-dimensional feature vectors. This already dominant approach was recently spurred even further, by the discovery that neural networks are capable of producing excellent high-dimensional embeddings of complex data domains [20, 21, 17]. To deal computationally with those rich and large embeddings, it has become a common practice to first compress them in a way that allows distances between data points to be approximated

from the compressed representations. This approach has grown into a major area of research, yielding many empirically successful and widely used algorithms, largely known as *quantization*.

This practice is tightly related to the phenomenon of *Euclidean dimension reduction*, discovered in the cornerstone work of Johnson and Lindenstrauss (1984) [16]. It states that, while the exact distances between n points may require up to $n - 1$ dimensions to represent, *approximate* distances require only $O(\log n)$ dimensions. This fundamental fact has far-reaching implications in theory and is manifestly useful in practice. In particular, it lays formal foundations for metric compression, by proving that dramatic compression is possible with almost no loss in the distance accuracy.

However, the most empirically successful quantization algorithms go *beyond* dimension reduction: rather than representing points as points of lower dimension, they design more general and sophisticated bit representations, tailored to the dataset, often called “hashes” or “codes” [24, 23]. This step had no rigorous justification, leading to the following question: *Can quantization provably improve over dimension reduction in compressing Euclidean distances?* My core thesis research answers this question affirmatively, with the following results:

- **Tight bounds:** In [6, 7], we obtain tight bounds on the bit complexity of representing all distances in an n -point Euclidean space up to distortion $1 \pm \varepsilon$. While the dimension reduction theorem states that each point can be represented by $O(\varepsilon^{-2} \log n)$ *coordinates* (each occupying a machine word), we show it can be represented by that many *bits*. Since the Johnson-Lindenstrauss theorem is tight [10, 18] – i.e., lower dimension is not possible – this establishes a rigorous separation between dimension reduction and quantization.

Perhaps surprisingly, the same algorithm also achieves the optimal compression bound for *general metric spaces*, settling their bit representation size in the low distortion regime. This ties to a long and prominent line of research in theoretical computer science on *distances oracles* [11, 12, 14, 22, 19], which studies the same problem in higher distortion regimes.

- **Nearest neighbor search:** In [5], we extend our technique to support distances to *query points* that had not been in the dataset, with nearly the same bit complexity. This enables approximate nearest neighbor search queries, yielding the smallest known data structure for this problem.
- **Practical algorithm:** There is often a divide between theoretical and practical algorithms, with many of the former ones – despite their strong formal guarantees – considered unsuitable for implementation. Bridging this gap is among the prime challenges toward developing new powerful tools for big data. We address this in [4], by presenting a variant of our algorithm that retains a worst-case near-optimal compression rate, and at the same time, empirically matches or improves over state-of-the-art heuristic quantization algorithms on standard benchmark datasets.

Fast Algorithms via Metric Compression

Compact metric representations are useful not only for saving space, but also for directly improving running times, as demonstrated in the following works.

Streaming semi-supervised learning. Semi-supervised learning is a branch of machine learning whose goal is to learn from a combination of labeled and unlabeled data. The influential work of Zhu et al. [25] cast the problem as label propagation on a graph, where labels spread from labeled to unlabeled nodes by a random walk, a fundamental process for measuring distance and similarity in a graph. This approach, while manifestly successful, entails a massive computation over the entire graph. A major challenge had been to apply this method to data streams, where data is exposed gradually and the stream is too large to store in full. In [9] we presented an algorithm that achieves this goal. Our method is based on a compression technique that shrinks the graph in stages while

provably preserving the result of the random-walk based learning process. Our experiments show high-accuracy classification results on real-world fast-moving streams of time-series medical signals and camera video feeds.

Fair clustering. The increasing societal impact of machine learning systems entails a new set of ethically and legally imposed constraints. A basic example is *fair clustering*, where one is required to minimize a clustering objective (e.g., k -median) while also ensuring minimal representation of a protected group (say, an ethnic or gender minority) in each cluster. The problem was defined in [13], but unfortunately, all existing algorithms had at least quadratic running time and could not scale to large datasets. In [1], we gave a near-linear time approximation algorithm for this problem, with worst-case guarantees, and were able to run it on datasets with millions of points.

Learning space partitions (Neural LSH). Locality-sensitive hashing (LSH), introduced by Indyk and Motwani [15], is classic paradigm for nearest neighbor search in sub-linear query time. The idea is to use a partition of the search space for hashing the dataset, enabling quick access to data points near any given query point. Much of the effort in this line of research is concentrated on finding better partitions. In [3], we show how to *learn* a partition for a given dataset, by utilizing the strong generalization power of modern machine learning tools, particularly neural networks.

Future Directions

- **Unifying theory and practice in Big Data:** A major open-ended direction in big data analysis is to integrate theoretical and practical advances into a well-founded and actionable discipline. Efforts in this vein can proceed in several concurrent directions: One is to adapt theoretical algorithms for implementation, by replacing their impractical components with practical (and possibly sub-optimal) analogs, while retaining their analytic core (as we have done in [4]). The other is to put empirically successful heuristics into analytic frameworks (as we have done in [8]), that could lead to better understanding, improvements, and perhaps the discovery of new analytic techniques. Ultimately, the goal is to smoothly translate theoretical advances into practical impact.
- **Learning on data streams:** Despite its prominence in applications, streaming data appears to have benefited less from the rapid quality surge of machine learning systems. Costly repeated passes over the full dataset, which are typically required in training, are unsuitable when the data accumulates gradually and each incoming point requires immediate response. Nonetheless, it is worthwhile to explore to what extent the power of batch machine learning can be adapted to the streaming setting. We have demonstrated this in [9], by adapting a classic learning paradigm that had been markedly known to be “transductive”, i.e., geared toward the batch setting. I believe there is potential in similar efforts for other streaming problems.
- **Learning-based algorithm design:** A recent and exciting line of work explores the use of machine learning in algorithm design (including our aforementioned work [3]). One reason to see promise in this direction is the salient ability of learning architectures (especially neural networks) to adapt to a given dataset. This could alleviate the need of classic algorithms to “plan for” the possibility of worst-case data at the cost of decreased performance on typical data – and thus play a key role in developing algorithms that are both provable and practical.

My ongoing research includes improving analytic bounds on practical approximation algorithms for Optimal Transport (continuing our recent initial effort in [2]), and exploring learning-based algorithms for numerical matrix approximations. Moving forward, I plan to continue developing the directions outlined above toward novel and impactful algorithms for emerging scalability challenges in data science and machine learning.

My Publications

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- [2] Arturs Backurs, Yihe Dong, Piotr Indyk, Ilya Razenshteyn, and Tal Wagner. “Scalable Nearest Neighbor Search for Optimal Transport”. *NeurIPS workshop on Optimal Transport in Machine Learning (preliminary version)* (2019).
- [3] Yihe Dong, Piotr Indyk, Ilya Razenshteyn, and Tal Wagner. “Learning Space Partitions for Nearest Neighbor Search”. *International Conference on Learning Representations (ICLR)* (2020).
- [4] Piotr Indyk, Ilya Razenshteyn, and Tal Wagner. “Practical data-dependent metric compression with provable guarantees”. *Advances in Neural Information Processing Systems (NeurIPS)*. 2017, pp. 2617–2626.
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- [9] Tal Wagner, Sudipto Guha, Shiva Kasiviswanathan, and Nina Mishra. “Semi-supervised learning on data streams via temporal label propagation”. *International Conference on Machine Learning (ICML)*. 2018, pp. 5095–5104.

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- [20] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, and Jeff Dean. “Distributed representations of words and phrases and their compositionality”. *NeurIPS*. 2013.
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